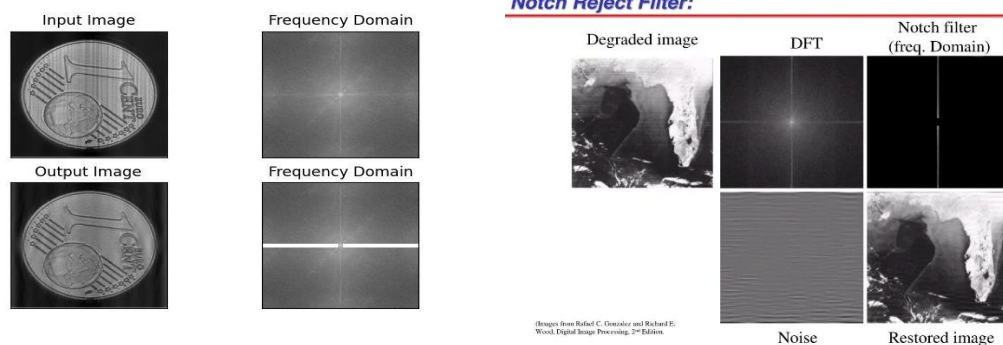


Question 1 – Manufacturing Quality Inspection



Scenario

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

(a) Interpretation of the Scenario

In modern manufacturing industries, cameras continuously inspect metal components to identify defects such as cracks, scratches, dents, or surface irregularities. During image acquisition, electrical interference from nearby machines or power lines introduces repetitive horizontal stripes across the captured images. This interference is known as Periodic Noise because it appears in a repeated pattern throughout the image. Therefore, removing periodic noise is essential to improve inspection accuracy and maintain product quality.

(b) Effect of Noise on Inspection Accuracy

Periodic noise significantly reduces the performance of automated inspection systems.

- It hides small cracks and scratches on the metal surface.
- The repeated stripes confuse edge detection algorithms.
- Image contrast decreases, making defects difficult to distinguish.
- The system may incorrectly classify defective products as good products.
- False alarms may also occur, increasing production cost.
- Manual inspection becomes necessary, reducing automation efficiency.

Hence, periodic noise directly affects the reliability and accuracy of quality inspection.

(c) Recommended Enhancement Technique

The most suitable enhancement technique is the Fourier Transform with Notch Filtering.

Working Process:

1. Convert the noisy image into the frequency domain using the Fourier Transform.
2. Detect bright frequency peaks caused by periodic noise.
3. Apply a Notch Filter to remove only the unwanted frequency components.
4. Perform the Inverse Fourier Transform to reconstruct the cleaned image.

This method removes repetitive stripe noise while preserving the important details of the metal component.

(d) Justification

The Fourier Transform separates image information into different frequency components.

Periodic noise appears as distinct peaks in the frequency spectrum.

A Notch Filter selectively removes only those unwanted frequencies while preserving useful image information.

In contrast, Notch Filtering removes only the periodic interference without affecting important image features. Therefore, it is the most suitable technique for manufacturing quality inspection.

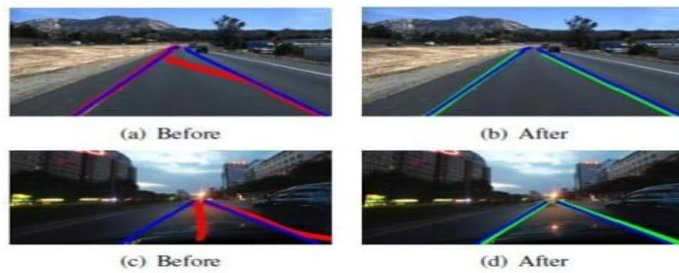
(e) Logical Sequence

1. Capture the image of the metal component.
2. Detect horizontal stripe interference.
3. Apply Fourier Transform.
4. Identify periodic noise frequencies.
5. Use Notch Filter to remove the unwanted frequencies.
6. Perform Inverse Fourier Transform.

Conclusion

Periodic noise caused by electrical interference can greatly reduce manufacturing inspection accuracy. Fourier Transform combined with Notch Filtering effectively removes repetitive stripe noise while preserving important defect information. This improves image quality, enhances defect detection, reduces false results, and increases the overall efficiency of automated quality inspection systems.

Question 2 – Autonomous Vehicle Navigation



Scenario

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

(a) Interpretation of the Scenario

Autonomous vehicles rely on cameras to recognize lane markings, road boundaries, pedestrians, traffic signs, and obstacles. During heavy rain, water droplets and environmental conditions introduce noise into captured images. However, the vehicle must still clearly detect lane markings and road edges for safe navigation. Therefore, the challenge is to remove rain-induced noise while preserving important edges that guide vehicle movement.

(b) Importance of Preserving Edges

Edges represent important boundaries in an image.

- Lane markings are identified using edges.
- Road boundaries define the driving path.
- Object detection depends on clear edges.
- Traffic sign recognition requires sharp boundaries.
- Blurred edges may lead to incorrect steering decisions.
- Preserving edges improves vehicle safety and navigation accuracy.

Hence, edge preservation is one of the most critical requirements in autonomous driving.

(c) Recommended Filtering Technique

The most suitable filtering technique is the Bilateral Filter.

Working Process:

1. Read the noisy road image.

2. Compare neighboring pixels based on distance.
3. Compare pixel intensity differences.
4. Smooth only similar pixels.

(d) Justification

Unlike Gaussian or Mean Filters, the Bilateral Filter considers both spatial distance and intensity difference between pixels.

This allows the filter to smooth flat regions while protecting object boundaries.

As a result:

- Lane markings remain sharp.
- Road edges remain clear.
- Traffic signs remain visible.
- Noise caused by rain is reduced.
- Navigation becomes safer and more accurate.

Therefore, Bilateral Filtering is the preferred choice for autonomous vehicle vision systems.

(e) Logical Sequence

1. Capture the road image.
2. Detect rain-induced noise.
3. Apply Bilateral Filter.
4. Remove unwanted noise.
5. Preserve lane edges.
6. Detect road boundaries.
7. Identify vehicles and obstacles.
8. Perform lane detection.

Conclusion

Heavy rain introduces noise into road images, making autonomous driving more challenging. The Bilateral Filter effectively removes noise while preserving important edges such as lane markings and road boundaries. This improves image quality, enhances object detection, increases navigation accuracy, and ensures safer driving in adverse weather conditions.